SYSTEMS, EACH WITH ITS CHOSEN SOLUTION AND SETTINGS

site: Grand Rapids, Kent County, Michigan: Michigan Residential Code figures, the NOAA Grand Rapids normals and the PVGIS October-to-April sun on each vertical side (solution grand-rapids-mi)
foundation: posts-at-corners (solution posts-at-corners)
foundation.embedment_mm = 1219 mm; no reason given
foundation.above_grade_mm = not given mm; reason: the sketch's '?' from grade to the floor; Jim's word gives it
foundation.section_mm = not given mm; reason: the post stock gives it: post_stock is the catalog's 6x6 UC4B post, dressed 5-1/2 in (wt.dressed_sizes); a number here overrides it
floor: platform (solution platform)
floor.width_mm = 2438 mm; no reason given
floor.depth_mm = 2438 mm; no reason given
floor.thickness_mm = not given mm; reason: the panel gives it: panel is the catalog's 4-1/2 in PRS 610 SIP (sip.prs610_panel_thickness); Jim's floor build-up may override it
floor.elevation_mm = not given mm; reason: the construction domain lays the floor's top at the datum, so a null here lies the panel its thickness below it
walls: walls-on-edges (solution walls-on-edges)
walls.height_mm = 3658 mm; reason: reading: the south elevation's '12' as wall height, floor top to roof underside; 12 ft overall above grade is the other reading
walls.thickness_mm = not given mm; reason: the panel gives it: panel is the catalog's 4-1/2 in PRS 610 SIP (sip.prs610_panel_thickness); a number here overrides it
walls.panel_width_mm = 1219 mm; reason: 4 ft is a sold width (sip.stock.premier_widths, sip.stock.acme_sizes); chosen narrow for handling and take-down; open for ranking
roof: flat-roof-over (solution flat-roof-over)
roof.south_eave_mm = 2438 mm; reason: reading: the hatched hangover on the south, 8 ft past the floor; centred, 5 ft each end, is the other reading
roof.east_eave_mm = 610 mm; reason: the sketch's 12 ft roof width less the 8 ft floor and the 2 ft west eave
roof.west_eave_mm = 610 mm; reason: the sketch's 2 ft west eave
roof.north_eave_mm = 610 mm; reason: the sketch's 2 ft north eave
roof.thickness_mm = not given mm; reason: the roof panel gives it: roof_panel is the catalog's 8-1/4 in Premier SIP, the thinnest listed whose R-value reaches the envelope's R-30 (sip.premier_thicknesses); a number here overrides it
glazing: Openings in a grid of rows stacked over the sill and columns side by side across one wall, filling the glazed height and the wall's inside length (solution opening-grid)
glazing.sides = south compass_sides; reason: Jim, 2026-10-18: only south-facing glass; no east, west or north window
glazing.first_sill_mm = 1219 mm; reason: Jim: the first 4 ft of the south wall is a SIP panel below the windows
glazing.glazed_height_mm = 2438 mm; reason: Jim: 8 ft of glass over the knee wall
glazing.rows = 2 count; reason: Jim: two stacked 4x8 windows, the upper one lighting the sleeping loft; one row (one 8x8) or two rows of two columns (four 4x4) fill the same 8 ft
glazing.columns = 1 count; reason: Jim: each window spans the south wall's 8 ft
weather: conditions (solution conditions)
weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side
finish: uniform (solution uniform)
finish.mean_reflectance = 0.5 ratio; reason: assumed mid-grey interior; Jim's wall finish gives it
finish.sky_angle_degrees = 65 degree; reason: assumed open sky; the site gives it
heating: wood-boiler-hydronic (solution wood-boiler-hydronic)
fuel: A cord (128 ft3 stacked) of seasoned hardwood, from the Michigan figure up to hickory (solution mixed-hardwood-cord)
doors: door-on-side (solution door-on-side)
doors.width_mm = 1219 mm; reason: Jim, 2026-10-18: the north door's framing is 4 ft wide by 8 ft tall
doors.height_mm = 2438 mm; reason: Jim: 8 ft tall door framing; a 4 ft tall 8 ft wide SIP panel sits over the door and its neighbour
doors.sides = north compass_sides; no reason given
doors.offset_mm = 0 mm; reason: assumed: the west end of the north wall; which end the door is at is open
loft_floor: A deck filling the floor inside the walls at an elevation, laid from the stock deck sheet (solution deck-inside-walls)
loft_floor.elevation_mm = 2438 mm; reason: Jim: the sleeping loft floor is 8 ft above the living floor
loft_floor.thickness_mm = not given mm; reason: the deck sheet gives it: deck_panel is the catalog's 23/32 in AdvanTech subfloor (wd.advantech_23_32_commerce), the IRC subfloor table's panel for a 24 in span (bom.subfloor_irc); a number here overrides it
joists: joist-framing (solution joist-framing)
joists.member_width_mm = not given mm; reason: the joist stock gives it: joist_stock is the catalog's dressed 2x6 (wt.dressed_sizes); a number here overrides it
joists.member_depth_mm = not given mm; reason: the joist stock gives it: joist_stock is the catalog's dressed 2x6 (wt.dressed_sizes); a number here overrides it
joists.spacing_mm = 406 mm; reason: assumed: 16 in on centre
joists.elevation_mm = not given mm; reason: the structural domain sets the joists' tops at the loft deck's elevation, so a null here takes it
joists.bears_on = west, east compass_sides; reason: the south wall is glass from 4 ft up and the north wall's door rises to 8 ft, so the joists can only land on the solid west and east walls
ledgers: A ledger along each wall the joists bear on, fastened to its face, the joists resting on its top (solution ledgers-under-joists)
ledgers.member_width_mm = not given mm; reason: assumed: a ledger cut from the joist stock, fastened to the wall's inside face; the fastening is not checked
ledgers.member_depth_mm = not given mm; reason: assumed: a ledger cut from the joist stock, as deep as the joists it carries
house: The building as the parts that bear on each other (solution load-bearing-parts)
panel: 4-1/2 in SIP to ANSI/APA PRS 610.1: EPS core between two 7/16 OSB facings (solution prs-610-sip-4-1-2)
lumber: Spruce-Pine-Fir No. 2 dimension lumber, NDS 2018 Table 4A (solution spf-no2)
joist_stock: A 2x6, dressed to 1-1/2 by 5-1/2 in (solution dressed-2x6)
header: Header of 1-3/4 by 11-7/8 in LVL plies, Microllam 2.0E-2600Fb (solution header-lvl-11-7-8)
header.plies = 2 count; reason: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads: The ground snow and the roof's dead load, on the walls the roof bears on (solution ground-snow-on-roof)
roof_loads.roof_dead_load_psf = 15 psf; reason: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio; reason: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = south, north compass_sides; reason: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls
post_stock: 6x6 pressure-treated post to UC4B, dressed 5-1/2 in square (solution post-6x6-uc4b)
roof_panel: Premier 8-1/4 in EPS SIP, R-30 at its lowest listed value; facings taken as 7/16 OSB, the thinnest ESR-4524 allows, and its Type I core at 1 pcf (solution premier-sip-8-1-4)
deck_panel: Huber AdvanTech 23/32 in OSB subfloor, 4 by 8 ft, Commerce GA mill weight (solution advantech-23-32)
window_glass: 1 in insulating glass unit of two 1/4 in lites and a 1/2 in airspace, Vitro TD-140 (solution igu-1in-two-quarter-lites)
door_slab: Therma-Tru Smooth-Star 1-lite fiberglass slab, 36 by 80 in, 1-3/4 in thick, as one retailer lists it (solution door-thermatru-smooth-star)
labour: Published, inferred and assumed hours for each task of the build, each naming its row (solution published-rates)
prices: Listed prices: a 6x6x8 treated post, an 8 ft 2x6, a 4x8 T&G subfloor sheet, a fiberglass entry door, picture windows by face area, and a wood boiler; each range's low and high rows are in rows (solution listed-prices)
sip_blanks: ACME Panel's listed SIP blanks, 7/16 in OSB both sides on EPS, priced from half to twice the list as no Michigan maker publishes one (solution acme-panel-blanks)
build_sequence: Post holes and posts, then SIP floor, walls and roof panels, the openings, a joisted loft deck and the heater (solution sip-shell-on-posts)

SOLVERS, EACH ANSWER WITH WHAT IT RESTS ON

average-daylight-factor (light): known
daylight_factor_percent = 7.69
sources: wd.vitro_td140_igu_65psf (published); ev.igu.viracon_u (published); ev.igu.viracon_shgc (published); ev.igu.viracon_vt (published)
glazing-balance (thermal): known
heat_loss_watt = 361.13
solar_gain_watt = 1931.75
sources: wd.vitro_td140_igu_65psf (published); ev.igu.viracon_u (published); ev.igu.viracon_shgc (published); ev.igu.viracon_vt (published)
design-day-load (thermal): known
loss_btu_per_hour = 2338.04
solar_gain_btu_per_hour = 6591.4
sources: sip.prs610_panel_thickness (published); sip.prs610_core_thickness (published); osb.gp_7_16 (published); sip.sipa_definition (published); wt.syn_osb_sheet (inferred); e1295.core (published); ev.sip_insulspan_4half_eps_r (published); sip.premier_thicknesses (published); pd.esr4524_facing (published); osb.gp_7_16 (inferred); pd.esr4524_core (published); e1295.core (inferred); wd.door_thermatru_smoothstar_120 (published); ev.door_thermatru_fc_ss_opaque_u (published); wd.vitro_td140_igu_65psf (published); ev.igu.viracon_u (published); ev.igu.viracon_shgc (published); ev.igu.viracon_vt (published)
heating-season (thermal): known
season_loss_btu = 10921786.36

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 1 of 3)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
31254d4	we 5af4c5f; rendered with we c3542b1	not to scale	we geometry